Early Structure-Function Coupling and Its Association with Myelination and EI Balance



PRESENTER:
Jiarui Mao
jiaruim@ad.unc.edu

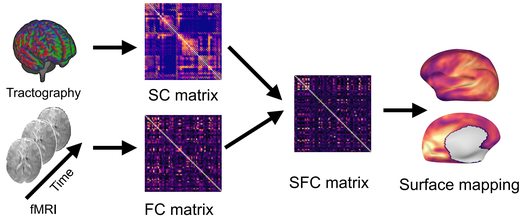
Background & Challenges

- Both structural and functional connectivity development is modular and segregated.
- Structure–function coupling (SFC) quantifies the dependence of SC and FC and varies systematically along the sensorimotor–association (S–A) axis.
- Vertex-level SFC in 0-5 years old infants have not yet be fully understood.

Aims

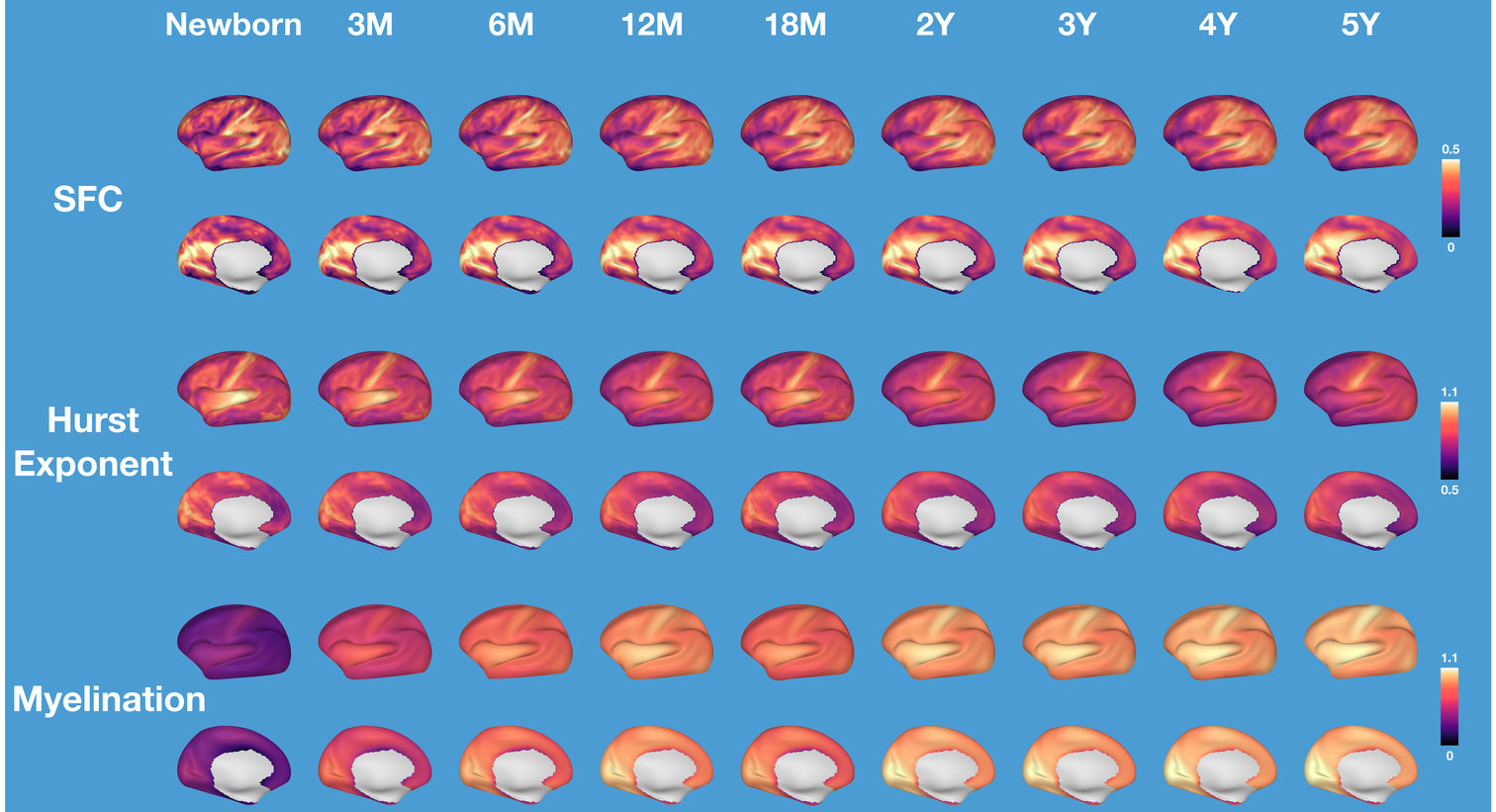
- To examine SFC in relation to cortical functional hierarchies (e.g., excitation-inhibition balance) and structural maturation (e.g., myelination) along S-A axis.
- Chart SFC development during the first 5 years of life

Materials & Methods

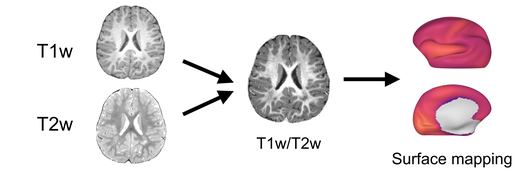


- 6-shell diffusion MRI, 382 scans from 217 subjects
- Resting-state fMRI, 1095 scans from 557 subjects processed with infant-centric pipeline
- Generalized Additive Mixed Model (GAMM) for longitudinal modelling

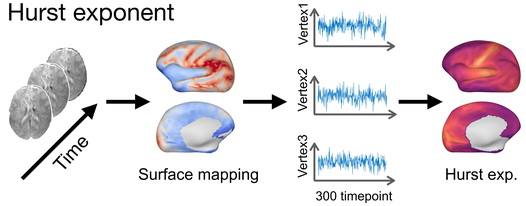
Structural functional coupling (SFC) develops along **Sensorimotor-Association (S-A) axis** in parallel with **myelination** and **E-I balance** during **the first 5 years of life.**



Myelination

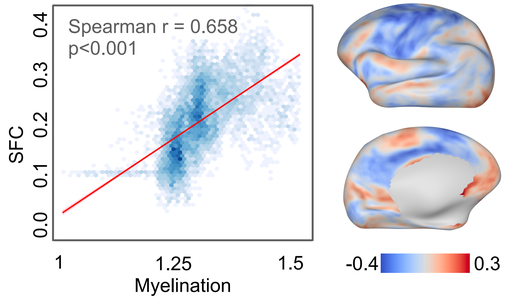


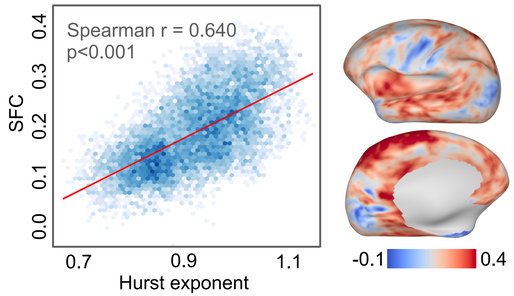
Hurst exponent



Results

- SFC increases with age, highest in visual and somatomotor network, weaker in high-order association networks.
- Hurst exponent and myelination decrease along the unimodal-to-transmodal gradient.





Jiarui Mao, Khoi Huynh, Kim-Han Thung, Sahar Ahmad, and Pew-Thian Yap

NIH EB008374 MH125479 MH133836
BRIC BIOMEDICAL RESEARCH IMAGING CENTER
UNC SCHOOL OF MEDICINE